# ELECTRICAL INSTALLATION CERTIFICATE

**Requirements for Electrical Installations - BS 7671: 2018+A2:2022**
**(IET Wiring Regulations 18th Edition)**

## Guidance for recipients:

This safety Certificate has been issued to confirm that the electrical installation work to which it relates has been designed, constructed, inspected and tested in accordance with BS 7671 (the IET Wiring Regulations).

You should have received an 'original' Certificate and the person that issued the Certificate should have retained a duplicate.

If you were the person ordering this work, but not the owner of the installation, you should pass this Certificate, or a full copy of it, immediately to the owner. The original Certificate is to be retained in a safe place and be shown to any person inspecting or undertaking work on the electrical installation in the future.

If you later vacate the property, this Certificate will demonstrate to the new owner that the electrical installation complied with the requirements of BS 7671 at the time the Certificate was issued.

The Construction (Design and Management) Regulations require that, for a project covered by those Regulations, a copy of this certificate, together with schedules, is included in the project health and safety document.

For safety reasons, the electrical installation will need to be re-inspected at appropriate intervals by a skilled person or persons, competent in such work. The maximum time interval recommended before the next inspection is stated in Section 3 under "NEXT INSPECTION".

This Certificate is intended to be issued only for a new electrical installation or for new work associated with an addition or alteration to an existing installation. It should not have been issued for the inspection and testing of an existing electrical installation. An "Electrical Installation Condition Report" should be issued for such an inspection.

This Certificate is only valid if the Schedule of Inspections has been completed to confirm that all relevant inspections have been carried out and where accompanied by Schedule(s) of Circuit Details and Test Results.

Where the installation includes a residual current device (RCD) it should be tested six-monthly by pressing the button marked 'T' or 'Test'. The device should switch off the supply and should then be switched on to restore the supply. If the device does not switch off the supply when the button is pressed, seek expert advice. For safety reasons it is important that this instruction is followed.

Where the installation includes an arc fault detection device (AFDD) having a manual test facility it should be tested six-monthly by pressing the test button. Where an AFDD has both a test button and automatic test function, manufacturer's instructions shall be followed with respect to test button operation.

Where the installation includes a surge protective device (SPD) the status indicator should be checked to confirm it is in operational condition in accordance with manufacturer's information. If the indication shows that the device is not operational, seek expert advice. For safety reasons it is important that this instruction is followed.

Where the installation includes alternative or additional sources of supply, warning notices should be found at the origin or meter position or, if remote from the origin, at the consumer unit or distribution board and at all points of isolation of all sources of supply.

# ELECTRICAL INSTALLATION CERTIFICATE
# [BS 7671: 2018+A2:2022 as amended]

*for Domestic and Similar Premises up to 100 A*

Requirements for Electrical Installations
BS7671 :2018+A2:2022 (IET Wiring Regulations 18th Edition)

**FT/EIC** 8789000001038

## Client Details

| | | | |
|---|---|---|---|
| Client | J T Consultancy | Installation | Parsons House |
| Address | 80-88 Collingdon Street<br>Luton | Address | Parsons Road<br>Washington<br>Tyne & Wear |
| Postcode | LU1 1RX | Postcode | NE37 1EZ |

## Details of the Installation

Description of premises Domestic ☑ Commercial ☐ Industrial ☐ Date of original installation: Not specified

Installation is New ☑ Addition ☐ Alteration ☐ Records Available Yes ☐ No ☑ RCD Risk assessment attached ☐

Description of the installation

Two bed apartment with kitchen, living room and bathroom.
DB found in hallway cupboard.

Extent of the installation covered by this certificate

Full Test and Inspection on property
100% Physical
100% Circuit
No Cavity
10% Void

Details of departures from BS 7671 (regulations 120.3, 133.1.3 and 133.5)

N/A

Details of permitted exception. (regulation 411.3.3) where applicable a suitable risk assessment(s) must be attached to this certificate

N/A

## Declaration for Design, Construction, Inspection and Testing (for sole person responsibility)

I being the person responsible for design, construction, inspection and the test of the electrical installation (as indicated by my signature below), particulars of which are described in Section 2, having exercised reasonable skill and care when carrying out the design, construction, inspection and test hereby CERTIFY that the design, construction, inspection and test for which i have been responsible is to the best of my knowledge and belief in accordance with BS 7671:2018, amended to 2022 except for the departures, if any, listed below. The extent of liability of the signatory or the signatories is limited to work described in Section 2 as subject of this certificate.

For the DESIGN / CONSTRUCTION / INSPECTION & TEST of the installation:

| | | | |
|---|---|---|---|
| Company | JTH LED Lighting Limited | Position | Electrical Supervisor |
| Inspector Name | James Harbord | Date | 30/11/2022 |
| Address | 6 Whitworth Road<br>Peterlee<br>SR8 2LY | Scheme No. | Branch No. |
| | | Signature | |

| | | | |
|---|---|---|---|
| Reviewed By | James Harbord | Reviewed By Signature | James Harbord |
| Reviewed By Date | 30/11/2022 | | |

**Next inspection I the designer recommend that this installation is further inspected after an interval of not more than** 10 years

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Requirements for Electrical Installations
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## Supply Characteristics and Earthing Arrangements

Earthing Arrangements TN-S ☐ TN-C-S ☑ TT ☐ Other ☐ If Other please specify: N/A

Number & Type of live conductors AC ☑ DC ☐ No. of phases: 1 No. of wires: 2

**Nature of Supply Parameters (Note: (1) by enquiry, (2) by enquiry or by measurement)**

Nominal voltage, $U/U_0$ (1): 230 v | Nominal frequency, $f$(1): 50 Hz | Confirmation of polarity ☑

Prospective fault current, $I_{pf}$ (2): 1.17 kA | External loop impedance, $Z_e$ (2): 0.20 Ω

Supply Protective Device BS (EN): 1361 HBC Type 2 | Type: 2 | Rated Current: 63 A

No. of Additional Supplies: N/A

## Particulars of Installation at the Origin

**Details of installation Earth Electrode** (where applicable) Type (e.g. rod(s), tape etc): | Location: | Electrode resistance to earth: Ω

**Means of Earthing**
Distributors facility ☑ Installation Earth Electrode ☐
Maximum Demand (load): 80 Amps ☑ KVA ☐

| Main Protective Conductors | Material | csa | | (✓) or Value | | (✓) or Value | |
|---|---|---|---|---|---|---|---|
| Earthing Conductor | Copper | 25 | mm² | Continuity Verified ☑ | Ω | Connection Verified ☑ | Ω |
| Protective Bonding Conductor | | | mm² | Continuity Verified ☐ | Ω | Connection Verified ☐ | Ω |

| | Material | csa | | (connection / continuity) | (✓) or Value | | (✓) or Value | |
|---|---|---|---|---|---|---|---|---|
| **Main Supply Conductor** | Copper | 25 | mm² | Water installation | NA | Ω | To structural steel ☑ | Ω |
| **Main Switch** Location: Ground Floor Meter Cupboard | | | | Gas installation pipes | NA | Ω | To lightning protection ☑ | Ω |
| | | | | Oil installation pipes | NA | Ω | Other | NA Ω |

**Fuse/device rating or setting** 63 A Voltage rating 230 V BS(EN) 60947-3 No. of Poles 2 Current Rating 100 A
**If RCD main switch:** Rated residual operating current I Δn mA Rated time delay ms Measured operating trip time ms

**Comments on existing installation** (in case of addition or alteration see section 644.1.2) use continuation sheet if needed

N/A

(For additions or alterations) cables concealed within trunking and conduits, or cables or conduits concealed under floors, in roof spaces and generally within the fabric of the building or underground may not have been inspected.

## Schedule of Inspection - Outcomes

| Indicates an inspection has been carried out and the result is satisfactory | ✓ | Indicates the inspection is not applicable to a particular item | N/A |
|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| 1.0 | Condition of consumer's intake equipment (visual inspection only) | ✓ | 8.0 | Circuits (Distribution and Final) | ✓ |
| 2.0 | Parallel or switched alternative sources of supply | N/A | 9.0 | Isolation and switching | ✓ |
| 3.0 | Protective measure: Automatic Disconnection of Supply (ADS) | ✓ | 10.0 | Current-using equipment (permanently connected) | ✓ |
| 4.0 | Basic Protection | ✓ | 11.0 | Identification and notices | ✓ |
| 5.0 | Protective measure other than ADS | ✓ | 12.0 | Location(s) containing a bath or shower | ✓ |
| 6.0 | Additional protection | ✓ | 13.0 | Other special installations or locations | ✓ |
| 7.0 | Distribution equipment | ✓ | 14.0 | Prosumer's low voltage electrical installation(s) | ✓ |

**SCHEDULES:** **This cerificate is only valid when** (enter quantities of schedules attached) 1 **schedules of circuit details and test results are attached**

Inspector's Name: James Harbord Signature: *James Harbord*

Date: 30/11/2022

# ELECTRICAL INSTALLATION CERTIFICATE - Circuit Details

**FT/EIC** 8789000001038

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| | | | |
|---|---|---|---|
| **Client Name** | J T Consultancy | **Installation Address** | Parsons House, Parsons Road , Washington, Tyne & Wear |
| **Client Address** | 80-88 Collingdon Street<br>Luton | **Postcode** | NE37 1EZ |
| **Client Postcode** | LU1 1RX | | |

**Distribution board details - Complete in every case**

SPD Details: Type(s)* T1 ☐ T2 ☐ T3† ☐ N/A ☑

Location: Hall Cupboard
Designation: DB 1
No. of ways: 12

**Complete only if the distribution board is not connected directly to the origin of the installation**

Overcurrent protective device for the distribution circuit: Supply to distribution board is from: Ground Floor Meter Cupboard

No. of phases: 1 BS(EN): 1361 HBC Type 2 Type: B Rating: 100 A

Nominal voltage: 230 V RCD BS(EN): 61008 Type: A Rating: 30 IΔn mA

## SCHEDULE OF CIRCUIT DETAILS

| Circuit No. and Line | Circuit designation | Type of wiring | Ref. method :j: | No. of points served | Circuit conductors csa (mm²) L / N | Circuit conductors csa (mm²) CPC | Maximum disconnection time (BS 7671) (S) | Overcurrent protective devices BS EN Number | Overcurrent protective devices Type No. | Overcurrent protective devices Rating (A) | Breaking capacity (KA) | BS 7671 Max. permitted Zs Other Other § 80% (Ω) | RCD BS EN Number | RCD Type No. | RCD IΔn (mA) | RCD Rating (A) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1/S | SOCKETS | A | 101 | 10 | 2.5 | 1.5 | 0.4 | 60898 MCB | B | 20 | 6 | 1.75 | 61008 | A | 30 | 100 |
| 2/S | SOCKETS | A | 101 | 9 | 2.5 | 1 | 0.4 | 60898 MCB | B | 20 | 6 | 1.75 | 61008 | A | 30 | 100 |
| 3/S | LIGHTING+MVHR UNIT | A | 101 | 19 | 1.5 | 1.5 | 0.4 | 60898 MCB | B | 6 | 6 | 5.82 | 61008 | A | 30 | 100 |
| 4/S | PANEL HEATER (BEDROOM) | A | 101 | 2 | 2.5 | 1.5 | 0.4 | 60898 MCB | B | 20 | 6 | 1.75 | 61008 | A | 30 | 100 |
| 5/S | WATER HEATER (BOOST) | A | 101 | 2 | 2.5 | 1.5 | 0.4 | 60898 MCB | B | 16 | 6 | 2.18 | 61008 | A | 30 | 100 |
| 6/S | SMOKES | A | 101 | 2 | 1.5 | 1.5 | 0.4 | 60898 MCB | B | 6 | 6 | 5.82 | 61008 | A | 30 | 100 |
| 7/S | HOB | A | 101 | 1 | 6 | 2.5 | 0.4 | 60898 MCB | B | 40 | 6 | 0.87 | 61008 | A | 30 | 100 |
| 8/S | KITCHEN SOCKETS | A | 101 | 4 | 2.5 | 1.5 | 0.4 | 60898 MCB | B | 32 | 6 | 1.09 | 61008 | A | 30 | 100 |
| 9/S | Oven | A | 101 | 1 | 6 | 2.5 | 0.4 | 60898 MCB | B | 32 | 6 | 1.09 | 61008 | A | 30 | 100 |
| 10/S | PANEL HEATER (HALL+ TOWEL RAIL) | A | 101 | 2 | 2.5 | 1.5 | 0.4 | 60898 MCB | B | 16 | 6 | 2.18 | 61008 | A | 30 | 100 |
| 11/S | PANEL HEATER (LIVING) | A | 101 | 2 | 2.5 | 1.5 | 0.4 | 60898 MCB | B | 16 | 6 | 2.18 | 61008 | A | 30 | 100 |
| 12/S | WATER HEATER (TIMER) | A | 101 | 2 | 2.5 | 1.5 | 0.4 | 60898 MCB | B | 16 | 6 | 2.18 | 61008 | A | 30 | 100 |

Wiring Types: **A** PVC/PVC, **B** PVC cables in metallic Conduit, **C** PVC cables in non-metallic Conduit, **D** PVC cables in metallic trunking, **E** PVC cables in non-metallic trunking, **F** PVC/SWA cables, **G** SWA/XPLE cables, **H** Mineral Insulated, **MW** Metal Work, **FM** Ferrous Metal, **O** Other

* SPD Type. Where a combined T1 + T2 or T2 + T3 device is installed, indicate by ticking both boxes.
t Where a T3 SPD is installed to protect sensitive equipment, enter Details of Circuits, of the Schedule of Test Results. (See Section 534 of BS 7671:2018+A2:2022.)
:j: See Table 4A2 of Appendix 4 of BS 7671:2018+A2:2022.
§ Where the maximum permitted earth fault loop impedance value stated in Max Zs column is taken from a source other than the tabulated values given in Chapter 41 of BS 7671:2018+A2:2022, state the source of the data in the appropriate cell for the circuit in the change to Schedule of Test Results

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| **Client Name** | J T Consultancy | | | **Installation Address** | Parsons House, Parsons Road , Washington, Tyne & Wear |
| **Client Address** | 80-88 Collingdon Street<br>Luton | **Client Postcode** | LU1 1RX | **Installation Postcode** | NE37 1EZ |

**Distribution board details - Complete in every case**

Location: Hall Cupboard
Designation: DB 1
No. of ways: 12 ☑ Supply polarity confirmed ☐ Phase sequence confirmed
No. of phases: 1 SPD: ☐ Operational status confirmed ☑ Not applicable

**Complete only if the distribution board is not connected directly to the origin of the installation**

Associated RCD (if any): BS (EN) 61008
$Z_{db}$ 0.28 Ω Operating at IΔn 30 ms
$I_{pf}$ 0.829 kA No. of poles 2 Time delay (if applicable)

## TEST RESULTS

| Circuit No. and Line | Circuit impedance Ω: Ring final circuits only r1 | rn | r2 | Fig 8 check (✓) | R1R2 or R2: R1 + R2 | R2 | Insulation resistance (Record lower reading): Test voltage V | L/L, L/N M(Ω) | L/E, N/E M(Ω) | Polarity | Max. Measured Zs (Ω) | RCD testing: All RCDs IΔn ms | Manual test button operation: RCD (✓) | AFDD (✓) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1/S | N/A | N/A | N/A | N/A | 0.19 | N/A | 500 | >299 | >299 | ✓ | 0.50 | 18 | ✓ | N/A |
| 2/S | N/A | N/A | N/A | N/A | 0.97 | N/A | 500 | >299 | >299 | ✓ | 1.24 | 18 | ✓ | N/A |
| 3/S | N/A | N/A | N/A | N/A | 1.20 | N/A | 500 | >299 | >299 | ✓ | 1.54 | 18 | ✓ | N/A |
| 4/S | N/A | N/A | N/A | N/A | 0.26 | N/A | 500 | >299 | >299 | ✓ | 0.57 | 18 | ✓ | N/A |
| 5/S | N/A | N/A | N/A | N/A | 0.09 | N/A | 500 | >299 | >299 | ✓ | 0.39 | 18 | ✓ | N/A |
| 6/S | N/A | N/A | N/A | N/A | 0.36 | N/A | 500 | >299 | >299 | ✓ | 0.65 | 18 | ✓ | N/A |
| 7/S | N/A | N/A | N/A | N/A | 0.07 | N/A | 500 | >299 | >299 | ✓ | 0.35 | 18 | ✓ | N/A |
| 8/S | N/A | N/A | N/A | N/A | 0.08 | N/A | 500 | >299 | >299 | ✓ | 0.33 | 17 | ✓ | N/A |
| 9/S | 0.35 | 0.36 | 0.69 | N/A | 0.25 | N/A | 500 | >299 | >299 | ✓ | 0.60 | 17 | ✓ | N/A |
| 10/S | N/A | N/A | N/A | N/A | 0.08 | N/A | 500 | >299 | >299 | ✓ | 0.34 | 17 | ✓ | N/A |
| 11/S | N/A | N/A | N/A | N/A | 0.29 | N/A | 500 | >299 | >299 | ✓ | 0.58 | 17 | ✓ | N/A |
| 12/S | N/A | N/A | N/A | N/A | 0.09 | N/A | 500 | >299 | >299 | ✓ | 0.38 | 17 | ✓ | N/A |

Details of circuits and/or installed equipment vulnerable to damage when testing

Date(s) dead testing 07/11/2022 To 07/11/2022
Date(s) live testing 07/11/2022 To 07/11/2022

Test instrument serial number(s)

Loop impedance 17080469 Insulation resistance 17080469 Continuity 17080469 RCD 17080469 E/Electrode N/A

Tested by: Name (capital letters) JAMES HARBORD Signature *James Harbord*

Position Electrical Supervisor Date 07/11/2022